

UDAY RAJ

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Education

Indian Institute Of Information Technology, Design and Manufacturing

Kurnool

Bachelor of Technology in Artificial Intelligence and Data Science - **Grade : 8.38**

2024 - 2028

Delhi Public School Sushant Lok

Gurugram

High School - (Physics, Chemistry, Mathematics) - **Grade : 95%**

2022 - 2024

Relevant Coursework

- Data Structures
- Algorithms Analysis
- Computer Vision
- Artificial Intelligence
- Machine Learning
- Deep Learning
- Statistical Analysis
- Discrete Mathematics

Experience

SVNIT Surat

Nov 2025 - Apr 2026

Research Summer Intern - Department of Artificial Intelligence

Surat, India

- Designed TransEvade under Dr. Praveen Kumar Chandaliya, a printability-constrained attention-hijacking attack for transformer-based Person Re-ID systems.
- Engineered a SegFormer/MiT Re-ID pipeline across Market-1501, DukeMTMC-reID, and MSMT17 using MiT-B0/B1 backbones, 256/512-D descriptors, and cosine retrieval.
- Optimized 500-1000 step adversarial pattern generation with Adam, ImageNet normalization, chest/waist/legs masks, and distract-suppress attention losses.
- Quantified 90-identity spatial ablations where torso patches reduced adversarial Rank-1 to 1.67%-7.04% on Duke/MSMT, exposing attention priors.

Projects

CourtSense AI - 3D Tennis Analytics Engine

[[GitHub](#)]

- Engineered a sports-CV pipeline turning tennis match footage into 150-frame 3D replay clips, streaming live JSON coordinates via FastAPI to a React + Three.js frontend.
- Fused 4 CV models – YOLOv8-Pose, SegFormer-B2 segmentation, OpenCV homography, and HSV/YOLO tracking – mapping 17 COCO keypoints onto court coordinates.
- Built a 6-stage physics pipeline (Kalman filtering, EMA smoothing at $\alpha = 0.3$, 5-frame rolling stats, Magnus-effect correction) for km/h speed and rpm spin estimates.

PaliGemma Vision-Language Model From Scratch

[[GitHub](#)]

- Rebuilt a PaliGemma-style VLM from scratch in PyTorch: a 12-layer SigLIP vision transformer encoding 224x224 images into 196 tokens (16x16 patches) feeding a Gemma decoder.
- Designed a 1.6M-parameter projector (768-to-2048-dim) merging image tokens at index 256000, with RMSNorm, RoPE, and grouped-query KV attention across an 8192-token context.
- Enabled KV-caching and causal masking for efficient autoregressive generation across 3 modular components – SigLIP encoder, projector, and Gemma decoder.

Technical Skills

Languages: Python, C, C++, SQL, TypeScript, HTML

Machine Learning Stack: PyTorch, TensorFlow, Keras, torchvision, scikit-learn, NumPy, Pandas

Computer Vision & Deep Learning: YOLOv8, SegFormer, ViT, SigLIP, OpenCV, Person Re-ID, Pose Estimation

Generative AI & LLMs: RoPE, RMSNorm, Grouped-Query Attention, KV-Cache, Beam Search, Multimodal AI

MLOps & Deployment: FastAPI, REST API, Docker, CUDA, Linux, Git, Jupyter, Quantization, Groq API

Achievements

- Placed global top 20 in the [[Shell.ai](#)] Hackathon among 1000+ submissions, optimizing ML model parameters.
- Qualified for **Amazon ML Summer School** – top 3,000 of 1,30,000+ applicants nationwide, clearing a 3-stage resume, SOP, and ML/DSA test process.
- Earned a 3-star rating of 1630 on [[CodeChef](#)] and Pupil (1308) on [[Codeforces](#)], and solved 100+ DSA problems on [[LeetCode](#)] and [[Code360](#)].

Positions of Responsibility

Google Developer Groups

Machine Learning Coordinator

Aug 2025 - Present

- Conducted a 12-week ML bootcamp for 200+ students, covering SGD, CNNs, and Kaggle workflows, resulting in 15+ student projects.

DataWorks Club

Computer Vision Coordinator

Aug 2025 - Present

- Mentored 100+ members through OpenCV and PyTorch workshops, building real-time CV pipelines with YOLOv8 and TensorRT targets of 30+ FPS.